Weak existence for adaptive porous flow with a negative drag jump

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Abstract

We prove weak existence for a stationary porous-flow model whose drag law changes at speed one and has a negative jump. At the threshold the drag is an arbitrary vector field, as in the free-transition formulation of the model. The result holds in every dimension $d \geq 2$ on bounded connected Lipschitz domains, for all finite growth exponents $r \geq 2$ and all the prescribed mixed pressure and flux data in their original Lebesgue and Sobolev spaces. The solution can be chosen with speed at least one almost everywhere. Finite divergence-free and gradient oscillations use the free unit sphere, while monotone auxiliary equations confine the only constitutive defects to bounded values. Comparing their exact solutions with the oscillatory targets gives summable corrections in the conjugate norms, so the argument also applies to rough data.

1 Introduction

Adaptive porous-flow models select different drag laws according to the local fluid speed. In the stationary mixed formulation, a drag field balances the pressure gradient and an external force, while the velocity satisfies mass conservation. Fumagalli and Patacchini introduced a model in which two radial laws are selected below and above a speed threshold [2]. Such a switch permits the constitutive response to change across the medium rather than fixing one law throughout the domain.

The sign of the drag jump changes the existence argument. The variational theory in [2, Theorems 4.10–4.11] treats the nonnegative jump in arbitrary dimension, whereas its negative-jump result is restricted to one dimension. The related monotone framework in [3, Theorem 3.3, Remark 3.4 and Corollary 4.5] makes the same distinction between general monotone laws and higher-dimensional nonmonotone existence. A negative jump destroys convexity of the switched dissipation. In several dimensions the mass constraint also permits nontrivial oscillations, so minimization cannot be transferred from the one-dimensional argument.

There is an important distinction in the original model: the drag at the threshold is completely free. Any vector L^s field may be used on the unit-speed set, not only a radial branch value or an energy subgradient. As observed in [2, Remark 3.6], weak solutions with this transition law need not be minimizers of the underlying energy. We address precisely that weak existence question, with the original mixed pressure and normal-flux conditions and with no extra boundedness assumption on the data.

Our result is that every such datum in dimension $d \geq 2$, with finite growth exponent $r \geq 2$ and conjugate exponent $s = r/(r-1)$, admits a velocity in L^r, a pressure in the prescribed affine $W^{1,s}$ space, and a drag in L^s satisfying the constitutive and weak mass equations. Moreover, the velocity can be chosen with speed at least one almost everywhere. Theorem 1 states the complete hypotheses. Its inputs and solution concept are those of the weak-existence question in [5]; the

extra speed bound is an output. Thus the theorem answers that question for every original datum, without requiring minimization of the switched energy.

The free unit sphere permits finite splits of an arbitrary velocity–drag pair into graph states, with divergence-free velocity differences and gradient drag differences. Near the graph these splits have small mixed moments in the two conjugate norms. Local finite-potential constructions of this kind are part of convex integration [4]. We give an explicit version on measurable sets that preserves the global pressure and both original boundary conditions.

The main difficulty is using such finite constructions with rough data: their bounds depend on the pair values, which need not be uniformly bounded. We first extend the high branch to a strongly monotone radial law, then add a bounded source multiplied by a velocity cutoff. Every resulting exact solution already lies on the high graph at large speeds; all its remaining defects occur in a bounded pair range. Finite oscillations can therefore reduce those defects. A uniform monotonicity comparison turns the approximate affine fields into exact solutions of another cutoff equation. Iterating with summable norm corrections gives a strong limit in the original spaces.

The proof uses the threshold's unrestricted drag essentially. It does not give a selected transition law, uniqueness, or an energy-minimizing solution. Section 2 gives the weak formulation and auxiliary equations; section 3 proves the finite constructions; section 4 completes the strong-limit argument. Equalities of fields are understood almost everywhere and divergence identities distributionally. Vector norms are Euclidean and $\|\cdot\|_p$ denotes the $L^p(\Omega)$ norm.

2 Problem and auxiliary equations

2.1 The weak problem

Let $d \ge 2$ and let $\Omega \subset \mathbb{R}^d$ be a bounded connected Lipschitz domain. Its boundary is partitioned, up to surface measure zero, into relatively open sets Σ_p and Σ_v. Fix

$$2 \le r < \infty, \qquad s = \frac{r}{r-1}. \tag{1}$$

The two coefficients $\phi_1 : [0,1] \to [0,\infty)$ and $\phi_2 : [1,\infty) \to [0,\infty)$ are continuous and nondecreasing. Assume that for some $c, C > 0$,

$$c a^{(r-2)/2} \le \phi_2(a) \le C(1 + a^{(r-2)/2}) \quad (a \ge 1), \qquad 0 < \phi_2(1) < \phi_1(1). \tag{2}$$

The negative jump is the last strict inequality. For $u \in L^r(\Omega;\mathbb{R}^d)$ the permitted drag fields form the set

$$\Lambda(u) = \left\{ \phi_1(|u|^2) u \mathbf{1}_{\{|u|<1\}} + \phi_2(|u|^2) u \mathbf{1}_{\{|u|>1\}} + h \mathbf{1}_{\{|u|=1\}} : h \in L^s(\Omega;\mathbb{R}^d) \right\}. \tag{3}$$

In particular the unit-speed drag is an arbitrary vector field in L^s. Each branch coefficient is evaluated only on its indicated speed region.

The data are

$$q \in L^r(\Omega), \qquad f \in L^s(\Omega;\mathbb{R}^d), \qquad u_0 \in L^r(\Sigma_v).$$

When Σ_p has positive surface measure, prescribe an extension $P \in W^{1,s}(\Omega)$ and set

$$V = \{\psi \in W^{1,s}(\Omega) : \operatorname{Tr}\psi = 0 \text{ on } \Sigma_p\}.$$

When $\Sigma_p = \varnothing$, use

$$P = 0, \qquad V = \left\{ \psi \in W^{1,s}(\Omega) : \int_\Omega \psi = 0 \right\}, \qquad \int_\Omega q = \int_{\partial\Omega} u_0. \tag{4}$$

These are the only cases: a nonempty relatively open part of a Lipschitz boundary has positive surface measure. Boundary integrals below use the Sobolev trace and surface measure.

A weak solution is a triple $u \in L^r(\Omega;\mathbb{R}^d)$, $p \in P + V$ and $z \in L^s(\Omega;\mathbb{R}^d)$ such that

$$z \in \Lambda(u), \qquad z = f - \nabla p, \qquad \int_\Omega (u) \cdot \nabla\psi = -\int_\Omega q\psi + \int_{\Sigma_v} u_0 \operatorname{Tr}\psi \quad (\psi \in V). \tag{5}$$

The velocity need not have a classical boundary trace; the last identity specifies the normal-flux condition at its actual regularity.

Theorem 1. *For every domain, boundary partition, coefficients and datum satisfying (1)–(4) and the conditions above, (5) has a weak solution. It can be chosen with $|u| \geq 1$ almost everywhere in Ω.*

The additional speed bound is a property of the constructed solution. It removes the low branch from the proof; it imposes no restriction on the data. The theorem concerns the free transition law (3). It does not assert uniqueness or minimization of the switched dissipation.

2.2 The affine constraints and a monotone extension

Write $X = L^r(\Omega;\mathbb{R}^d)$ and define

$$\begin{aligned} \ell(\psi) &= -\int_\Omega q\psi + \int_{\Sigma_v} u_0 \operatorname{Tr}\psi, \\ \mathcal{C} &= \{u \in X : \int_\Omega (u) \cdot \nabla\psi = \ell(\psi) \quad \forall \psi \in V\}, \\ K &= \{v \in X : \int_\Omega (v) \cdot \nabla\psi = 0 \quad \forall \psi \in V\}, \qquad Y = \nabla V \subset L^s(\Omega;\mathbb{R}^d). \end{aligned}$$

The following linear facts will recover pressure after the velocity has been constructed. They do not depend on either nonlinear branch.

Proposition 2 (Linear pressure and mass constraints)**.** *The gradient norm on V is equivalent to its $W^{1,s}$ norm. The set $\mathcal{C}$ is a nonempty weakly closed affine translate of K, and*

$$K^\perp = Y = \nabla V.$$

The gradient map $V \to Y$ has a bounded inverse, with Y closed in L^s.

Proof. These are the linear statements of [2, Lemmas 4.1–4.2, pp. 573–574], using the homogeneous pressure space defined on p. 572. The source's pressure space is our V and its velocity kernel is our K. Its operator $B(u)(\psi) = \int_\Omega (u) \cdot \nabla\psi$ induces an isomorphism $X/K \to V^*$, and its adjoint is an isomorphism $V \to K^\perp$. The Poincare and trace bounds on this space make $\ell \in V^*$, so $B(u) = \ell$ has a representative in X. The adjoint is exactly the gradient under L^r–L^s duality. Its image is closed because annihilation of each member of K passes through an L^s limit. Finally each defining gradient pairing for $\mathcal{C}$ is weakly continuous. The source's linear proof uses the trace constraints and boundary integrals only up to null sets, so the present boundary-partition convention gives the same statements. □

Set $\beta = \phi_2(1) > 0$ and extend the high law continuously below its threshold:

$$\alpha(t) = \begin{cases} \beta, & 0 \leq t \leq 1, \\ \phi_2(t^2), & t > 1, \end{cases} \qquad A(a) = \alpha(|a|)a \quad (a \in \mathbb{R}^d). \tag{6}$$

This extension permits exact monotone auxiliary equations for the original data, without changing the weak affine constraints.

Lemma 3. *There is $\gamma > 0$, depending only on β, c, r, such that for all $a, b \in \mathbb{R}^d$,*

$$(A(a) - A(b)) \cdot (a - b) \geq \beta |a - b|^2, \qquad (A(a) - A(b)) \cdot (a - b) \geq \gamma |a - b|^r. \tag{7}$$

Also $A : L^r(\Omega; \mathbb{R}^d) \to L^s(\Omega; \mathbb{R}^d)$ is continuous. More generally, if a_n, c_n have uniformly integrable rth powers and $\|a_n - c_n\|_r \to 0$, then $\|A(a_n) - A(c_n)\|_s \to 0$.

Proof. The coefficient $g = \alpha - \beta$ is nonnegative and nondecreasing. If $|a| \geq |b|$, then

$$(g(|a|)a - g(|b|)b) \cdot (a - b) = g(|b|)|a - b|^2 + (g(|a|) - g(|b|))(|a|^2 - a \cdot b) \geq 0.$$

This proves the quadratic bound. Put $k_0 = \min(\beta, c)$; $\alpha(t) \geq k_0 t^{r-2}$. Let $R_0 = \max(|a|, |b|)$ and $t_0 = \min(|a|, |b|)$. If $t_0 \geq R_0/2$, the same expansion gives

$$(A(a) - A(b)) \cdot (a - b) \geq k_0 (R_0/2)^{r-2} |a - b|^2 \geq k_0 2^{4-2r} |a - b|^r.$$

If $t_0 < R_0/2$, use $a \cdot b \leq R_0 t_0$ to obtain

$$\begin{aligned}(A(a) - A(b)) \cdot (a - b) &\geq (R_0 - t_0)(\alpha(R_0)R_0 - \alpha(t_0)t_0) \\ &\geq \alpha(R_0)(R_0 - t_0)^2 \geq (k_0/4)R_0^r \geq (k_0/4)(2/3)^r |a - b|^r.\end{aligned}$$

Thus one can take $\gamma = k_0 \min(2^{4-2r}, \frac{1}{4}(\frac{2}{3})^r)$. The case $a = b = 0$ is immediate; no division by a speed is required.

The growth assumptions imply

$$|A(a)|^s \leq C_A(1 + |a|^r). \tag{8}$$

For the stated families, bounded L^r norms make their large-value sets small in measure. On a fixed velocity ball A is uniformly continuous. Together with $a_n - c_n \to 0$ in measure this gives $A(a_n) - A(c_n) \to 0$ in measure. Their sth-power differences are uniformly integrable by (8), hence converge to zero in L^s. Indeed, split their integral into a set where the difference exceeds a fixed small number and its complement; the first set has vanishing measure and the second contribution is bounded by that number to the power s times $|\Omega|$. Strong L^r convergence makes the rth powers uniformly integrable, which proves continuity at a fixed field. To see the stability under small errors explicitly, write $c_n = a_n + e_n$ with $e_n \to 0$ in L^r. Then $|c_n|^r \leq 2^{r-1}(|a_n|^r + |e_n|^r)$. The total integrals of $|e_n|^r$ are arbitrarily small after a finite prefix, and the prefix has absolutely continuous integrals. This proves uniform integrability. In particular, it holds for a fixed rough field plus uniformly bounded increments and for all families close to it in L^r. □

Lemma 4. *For every $F \in L^s(\Omega; \mathbb{R}^d)$ there is a unique pair $u \in \mathcal{C}$, $p \in P + V$ satisfying $A(u) = F - \nabla p$.*

Proof. The convex C^1 function

$$J(a) = \begin{cases} \beta |a|^2/2, & |a| \leq 1, \\ \beta/2 + \displaystyle\int_1^{|a|} \phi_2(t^2) t \, dt, & |a| > 1 \end{cases}$$

has gradient A and bounds $(c/r)|a|^r - C_J \leq J(a) \leq C_J(1 + |a|^r)$. Minimize

$$\int_\Omega J(u) - \int_\Omega (F - \nabla P) \cdot u \quad \text{over } u \in \mathcal{C}.$$

Coercivity and Holder's inequality bound a minimizing sequence in L^r. Reflexivity and weak sequential compactness give a weak limit in $\mathcal{C}$; see [1, Theorems 3.18 and 4.10]. The supporting-plane inequality at this limit, $J(v) \geq J(u) + A(u) \cdot (v - u)$, proves weak lower semicontinuity after integration, since $A(u) \in L^s$. The minimum is therefore attained.

For $w \in K$, differentiating along $u + tw$ gives $\int_\Omega (A(u) - F + \nabla P) \cdot w = 0$. The segment identity for J and lemma 3 justify the derivative in the dual pairing; its integrand is bounded by $C(1 + |u|^{r-1} + |w|^{r-1})|w|$, which is integrable, so the exchange of the segment and spatial integrals is justified. By proposition 2 there is $v \in V$ with $\nabla v = F - \nabla P - A(u)$. Then $p = P + v$ has the required sign and trace convention. For two solutions, their velocity difference is in K and their pressure difference in V. Testing the equation difference by the velocity difference cancels pressure; (7) and proposition 2 give uniqueness. □

2.3 Cutoff equations and bounded defects

We will alter drag on bounded velocity sets while leaving large velocities on the high graph. For $R \geq 1$ use the $1/R$-Lipschitz function

$$\chi_R(a) = \begin{cases} 1, & |a| \leq R, \\ 2 - |a|/R, & R < |a| < 2R, \\ 0, & |a| \geq 2R. \end{cases} \tag{9}$$

For a bounded measurable source $b : \Omega \to \mathbb{R}^d$, set $F_{b,R}(x, a) = A(a) + b(x)\chi_R(a)$.

Lemma 5. *If $M = \|b\|_\infty$ and $M/R \leq \beta/4$, the original affine constraints have an exact solution*

$$u \in \mathcal{C}, \quad p \in P + V, \quad z = F_{b,R}(x, u) = f - \nabla p. \tag{10}$$

Moreover, if an affine pair $(\widetilde{u}, \widetilde{z})$ satisfies $\widetilde{z} = F_{b,R}(x, \widetilde{u}) + d$, $d \in L^s$, then

$$\|u - \widetilde{u}\|_r^{r-1} \leq \frac{2}{\gamma}\|d\|_s. \tag{11}$$

Here an affine pair means $\widetilde{u} \in \mathcal{C}$ and $\widetilde{z} = f - \nabla\widetilde{p}$ with $\widetilde{p} \in P + V$.

Proof. Average the two bounds in (7) and subtract the cutoff error $M|a - c|^2/R$:

$$(F_{b,R}(x, a) - F_{b,R}(x, c)) \cdot (a - c) \geq \frac{\gamma}{2}|a - c|^r + \frac{\beta}{4}|a - c|^2. \tag{12}$$

To obtain existence, for $v \in L^2$ let $T(v)$ be the velocity from lemma 4 with force $f - b\chi_R(v)$. This belongs to $L^r \subset L^2$. Comparing two such equations, the pressure term vanishes and the quadratic bound gives

$$\|T(v) - T(w)\|_2 \leq \frac{M}{\beta R}\|v - w\|_2 \leq \frac{1}{4}\|v - w\|_2.$$

Iteration is Cauchy in the complete space L^2; its limit is a fixed point by this Lipschitz bound. As a value of T it belongs to L^r, and its auxiliary pressure gives (10). For comparison, both velocity fields have the same affine mass data, so their drag difference pairs to zero with their velocity difference. Using (12) and Holder gives $(\gamma/2)\|u - \widetilde{u}\|_r^r \leq \|d\|_s\|u - \widetilde{u}\|_r$, proving (11), including the zero-difference case. □

The closed part of the original graph which we shall use is

$$\Gamma = \{(a, z) : |a| = 1,\ z \in \mathbb{R}^d\} \cup \{(a, A(a)) : |a| > 1\}. \tag{13}$$

Its continuous nonnegative defect is

$$G(a, z) = \begin{cases} 1 - |a|, & |a| \le 1, \\ \min(|a| - 1, |z - A(a)|), & |a| > 1. \end{cases} \tag{14}$$

Continuity at the sphere follows from $G \le \big||a| - 1\big|$ there, and $G^{-1}(0) = \Gamma$. For a solution of (10), the residual $z - A(u) = b\chi_R(u)$ is bounded and vanishes at $|u| \ge 2R$. Consequently

$$G(u, z) > 0 \implies |u| < 2R, \quad |z| \le \max_{|a| \le 2R} |A(a)| + M. \tag{15}$$

This is the boundedness used below. The solution itself may have unbounded velocity and drag.

3 Finite oscillatory constructions

3.1 Wave connections and the full finite hull

For pairs $U_i = (a_i, z_i) \in \mathbb{R}^d \times \mathbb{R}^d$ the relation

$$(a_1 - a_2) \cdot (z_1 - z_2) = 0 \tag{16}$$

permits a divergence-free velocity oscillation and a gradient drag oscillation with one common normal. Indeed there is a unit vector ξ such that $a_1 - a_2$ is perpendicular to ξ and $z_1 - z_2$ is parallel to ξ: use its direction if the drag difference is nonzero, and otherwise choose a perpendicular direction to the velocity difference. This uses $d \ge 2$.

A finite laminate here means a finite probability measure obtained from a single mean pair by finitely many binary splits $U = \theta U_1 + (1 - \theta)U_2$, $0 \le \theta \le 1$, each satisfying (16). Zero-weight states can be omitted. We first establish that every pair can be split into states of the graph (13).

Lemma 6. *Every pair $(a, z) \in \mathbb{R}^{2d}$ is the mean of a finite laminate supported in Γ. On each set $|a|, |z| \le D$ the atoms can be chosen in a uniformly bounded set, with uniformly bounded split depth.*

Proof. For $|a| = \rho \le 1$, write $a = \rho e$ and choose a unit vector $b \perp e$. At $a = 0$ choose any orthonormal e, b. The two unit states

$$(\rho e \pm \sqrt{1 - \rho^2}\, b, z)$$

with equal weights have mean (a, z) and satisfy (16). When $\rho = 1$ the original graph state suffices.

For $|a| > 1$, choose a unit $b \perp a$, put $\delta = A(a) - z$, and choose $T > |\delta| + |a \cdot \delta| + 1$. For $\sigma \in \{-1, 1\}$ set

$$z_\sigma = z + \sigma T b, \qquad e_\sigma = \frac{A(a) - z_\sigma}{|A(a) - z_\sigma|}, \qquad d_\sigma = a - e_\sigma.$$

The denominator is nonzero by the choice of T. Along the ray $y_\sigma(\lambda) = e_\sigma + \lambda d_\sigma$, $\lambda \ge 1$, the speed is at least $|a| > 1$ and $d_\sigma \cdot y_\sigma(\lambda) > 0$. Consider

$$F_\sigma(\lambda) = d_\sigma \cdot (A(y_\sigma(\lambda)) - z_\sigma).$$

At the initial point,

$$F_\sigma(1) = a \cdot \delta - |\delta - \sigma T b| < -1.$$

The quadratic bound in (7) gives, for $\lambda > \mu \geq 1$,

$$F_\sigma(\lambda) - F_\sigma(\mu) \geq \beta(\lambda - \mu)|d_\sigma|^2.$$

Thus the continuous function has a unique zero $\lambda_\sigma > 1$. Define

$$a^h_\sigma = e_\sigma + \lambda_\sigma d_\sigma, \qquad z^u_\sigma = \frac{\lambda_\sigma z_\sigma - A(a^h_\sigma)}{\lambda_\sigma - 1}.$$

The unit state (e_σ, z^u_σ) and the high state $(a^h_\sigma, A(a^h_\sigma))$, with weights $(\lambda_\sigma - 1)/\lambda_\sigma$ and $1/\lambda_\sigma$, have mean (a, z_σ). Their velocity difference is $-\lambda_\sigma d_\sigma$, their drag difference is $\lambda_\sigma(z_\sigma - A(a^h_\sigma))/(\lambda_\sigma - 1)$, and their scalar product is zero by the root equation. Finally (a, z_+) and (a, z_-) have the same velocity and mean (a, z) with equal weights. This gives a two-level laminate.

For the uniform bounds, choose T with a common strict margin on the compact input range, including its limiting sphere $|a| = 1$. Let D_δ bound $|\delta|$. The transverse component gives

$$|d_\sigma| \geq |e_\sigma \cdot b| \geq \frac{1}{T + D_\delta} =: m > 0.$$

If C_F bounds $|F_\sigma(1)|$, the preceding monotonicity gives $\lambda_\sigma \leq 1 + C_F/(\beta m^2)$. On this bounded range the parameters and λ form a compact set. Uniform continuity of F_σ, together with $F_\sigma(1) < -1$, keeps $\lambda_\sigma - 1$ uniformly positive. It therefore also bounds z^u_σ. The high atoms and the subunit construction are bounded as well. No continuous choice of the perpendicular vector is needed. □

3.2 Mixed moment bounds near the graph

Fullness of the finite hull alone does not give small corrections near Γ. The following estimate supplies that property in the two conjugate norms of the weak problem.

Lemma 7. *For every $D \geq 1$ there are finite B_D, C_D such that every $U = (a, z)$ with $|a|, |z| \leq D$ is the mean of a finite laminate ν_U supported in $\Gamma \cap \{|a'|, |z'| \leq B_D\}$, with*

$$\int_{\mathbb{R}^{2d}} \left(|a' - a|^r + |z' - z|^s\right) d\nu_U(a', z') \leq C_D G(a, z). \tag{17}$$

The number of atoms and split depth can be bounded independently of U.

Proof. Inside the unit ball, the construction in lemma 6 has velocity moment $(1-\rho^2)^{r/2} \leq 2^{r/2}(1-\rho)$ and zero drag moment.

For $a = (1 + \varepsilon)e$, $0 < \varepsilon < 1/4$, choose a unit $b \perp e$, a fixed $T > 0$, and put

$$\theta = 2\varepsilon, \qquad a_+ = 2e, \qquad a_- = \rho e, \quad \rho = \frac{1 - 3\varepsilon}{1 - 2\varepsilon}, \qquad z_+ = z + Tb, \quad z_- = z - \frac{\theta}{1 - \theta} Tb.$$

The weights $\theta, 1 - \theta$ give mean (a, z), and the two differences in this split are perpendicular. Replace the plus pair by lemma 6's laminate and the minus pair by two unit states with common drag z_-. The plus inputs stay in a fixed compact set, so their atoms have a uniform bound and contribute at most $C_D\varepsilon$. For each minus unit state,

$$|a' - a|^2 \leq 9\varepsilon^2 + 4\varepsilon, \qquad |z_- - z| \leq 4T\varepsilon.$$

Here $1-\rho=\varepsilon/(1-2\varepsilon)\le 2\varepsilon$. Its mixed moment is at most $C(\varepsilon^{r/2}+\varepsilon^{s})\le C\varepsilon$, since $r\ge 2$ and $s>1$. This gives a bound by $|a|-1$ near the sphere. When $|a|-1\ge 1/4$, the bounded atoms from lemma 6 give the same bound after enlargement of the constant.

For a second bound, suppose $|a|>1$ and $\delta=|z-A(a)|\in(0,1/2)$. Split the pair into the exact high state $(a,A(a))$ of weight $1-\delta$ and the mean pair

$$\left(a,A(a)+\frac{z-A(a)}{\delta}\right)$$

of weight δ. Their velocities agree, so the split is allowed. The second input stays bounded; replace it using lemma 6. The first atom has drag deviation δ, and the other bounded atoms have total weight δ. The mixed cost is at most $C_D\delta$. For $\delta\ge 1/2$ use the full bounded construction, and for $\delta=0$ use the exact graph state. Selecting the cheaper of this construction and the sphere construction proves (17). Each case uses a fixed finite binary tree, so the bounds on atom number and depth are uniform. □

Enlarge the compact-set constants to nondecreasing majorants. Choose a continuous nondecreasing $k:[1,\infty)\to[1,\infty)$ which dominates them with one unit of radius slack; interpolation of a nondecreasing sequence with $k(n)\ge C_{n+2}$ suffices. Set

$$H(a,z)=k(1+|a|+|z|)G(a,z). \tag{18}$$

Then $H\ge G$, its zero set is Γ, and at every pair one of the laminates has mixed cost at most $H(a,z)$. The function is bounded on compact pair sets. By (15), $H(u,z)$ is bounded for every cutoff solution, although no r-growth bound for H at arbitrary mismatched pairs is asserted.

3.3 Measurable masks and exact affine perturbations

The local finite-potential construction for laminates is familiar from convex integration; see, for example, [4, Proposition 4.5]. We give the required form directly, because the initial affine pair may be rough and the set on which it is changed is merely measurable.

Lemma 8. *Let $U=(u,z)$ be an affine pair, and let E be a measurable subset of a compact set contained in Ω, with U bounded on E. For every $\epsilon>0$ there are fields $W_n=(w_n,\omega_n)$ and affine pairs $\widetilde{W}_n=(\widetilde{w}_n,\widetilde{\omega}_n)$ such that*

$$W_n=U \text{ off } E,\quad W_n-U \text{ uniformly bounded}, \tag{19}$$

$$\int_E H(W_n)<\epsilon, \tag{20}$$

$$\limsup_n\left(\|w_n-u\|_r^r+\|\omega_n-z\|_s^s\right)\le\int_E H(U)+\epsilon, \tag{21}$$

$$\|\widetilde{w}_n-w_n\|_r+\|\widetilde{\omega}_n-\omega_n\|_s\longrightarrow 0. \tag{22}$$

The perturbations $\widetilde{W}_n-U$ are compactly supported divergence-free/gradient pairs. For this fixed construction they also have a uniform L^∞ bound.

Proof. Partition E into finitely many measurable value bins E_j, with constants C_j such that $|U-C_j|\le\delta$ on E_j. Here and below a pair norm is $|(a,z)|=|a|+|z|$. Choose $\delta\le 1$, so all centers remain in one fixed compact range. Choose the laminates of lemma 7 at those centers. Their atoms have a common bound independent of the bin precision. On a bin the ideal field takes values

$$U(x)+(a_{j,\ell},z_{j,\ell})-C_j,$$

where $(a_{j,\ell}, z_{j,\ell})$ is a graph atom. Uniform continuity of H on that compact range makes its integral on E smaller than ϵ by choosing δ small. It also makes $\sum_j |E_j| H(C_j) \le \int_E H(U) + \epsilon$ with any prescribed smaller error in place of ϵ.

We realize the state fractions while approximating these ideal fields by exact affine ones. For a binary split with difference $q = (q_u, q_z)$ and weights $\theta, 1-\theta$, choose a unit ξ with $q_u \cdot \xi = 0$ and $q_z = t\xi$. Let h be a periodic profile taking values $1-\theta$ and $-\theta$ on fractions θ and $1-\theta$ of each period. Its mean is zero, so it has a bounded periodic Lipschitz primitive L. For a measurable parent phase F, the ideal correction is $\mathbf{1}_F h(N\xi \cdot x) q$.

Approximate $\mathbf{1}_F$ by smooth amplitudes η, between zero and one, supported in a fixed compact interior neighborhood. Convolution gives convergence in both L^r and L^s; see [1, Theorem 4.22]. Set $S = q_u \otimes \xi - \xi \otimes q_u$, so S is skew and $S\xi = q_u$. The potentials

$$\Psi_N(x) = \frac{1}{N} S\eta(x) L(N\xi \cdot x), \qquad v_N(x) = \frac{t}{N} \eta(x) L(N\xi \cdot x) \tag{23}$$

give

$$(\operatorname{div} \Psi_N, \nabla v_N) = \eta h(N\xi \cdot x) q + \frac{L(N\xi \cdot x)}{N} (S\nabla\eta, t\nabla\eta).$$

Row divergence is used for the matrix. First choose the smooth amplitude with small mask error, and then choose N so the last term is small in L^∞. Differentiation is also valid distributionally: L is continuous and piecewise affine, so integration by parts on its affine intervals has cancelling interface terms. Fubini along the ξ coordinate gives the displayed derivative identity.

The velocity perturbation is divergence-free because the potential is skew. It pairs to zero with the gradient of every $\psi \in V$: first use smooth tests, then the local Sobolev approximation of [1, Theorem 9.2] on the compact support, and pass by Holder. The drag perturbation is a global gradient. Its pressure correction is $-v_N$, with the mean restored by a constant in the pure-flux case. Thus both original boundary conditions are preserved, also on domains with holes.

Periodic averaging gives the required fractions on every measurable parent phase. For a smooth compactly supported weight, integrate a zero-mean profile's bounded primitive in direction ξ, obtaining an error $O(N^{-1})$. Density in L^1 extends this to $\mathbf{1}_F$; see [1, Theorem 4.12] and mollification. Apply this also to each phase indicator minus its mean fraction.

Carry out the finite split trees in order. At index n, choose the outer frequencies with errors below $1/n$. Their ideal phase masks are then fixed measurable sets. For each inner split, smooth its parent mask with error below $1/n$ in both norms and choose its frequency so its fraction error and potential remainder are below $1/n$. There are only finitely many splits. Their free corrections sum to a uniformly bounded compactly supported perturbation, and the accumulated mask and remainder errors tend to zero. On the disjoint bins the corresponding ideal targets take the leaf values above, so they have a uniform bound on E.

The conditional phase measures converge to the prescribed weights. Consequently the ideal mixed increment cost converges to the sum of the bin measures times the laminate moments, bounded by $\sum_j |E_j| H(C_j)$. Choosing the bin error smaller than ϵ proves (21). The other properties follow from the construction. All bounds here are for a fixed finite collection of bins and trees; no measurable selection of a continuum of laminates is needed. □

4 Proof of existence

We now use the finite oscillations to improve exact solutions, keeping the original data fixed. Call a pair a compact-source cutoff solution if it satisfies (10), b vanishes outside a compact subset of Ω,

and $R \geq 1$, $\|b\|_\infty / R \leq \beta/4$. The zero source is allowed. Recall that H is the continuous weighted defect (18); it is integrable on every such solution by (15).

Lemma 9 (Exact improvement)**.** *For every compact-source cutoff solution $U = (u, z)$ and every $\eta > 0$ there is another one, $U' = (u', z')$, with the identical original data and*

$$\int_\Omega H(U') < \eta, \qquad \|u' - u\|_r^r + \|z' - z\|_s^s \leq 2 \int_\Omega H(U) + \eta. \tag{24}$$

Proof. Since $H(U)$ is bounded, choose a compact interior set K_0 with $\int_{\Omega \setminus K_0} H(U) < \eta/10$ and set $E = K_0 \cap \{G(U) > 0\}$. Compact interior exhaustion and dominated convergence justify the choice. The same bound holds outside E, and U is bounded on E by (15). Use lemma 8 with an error $\epsilon < \eta/10$, small enough that the target H integral on E is also below $\eta/10$. Write its ideal targets $W_n = (w_n, \omega_n)$ and affine approximants $\widetilde{W}_n = (\widetilde{w}_n, \widetilde{\omega}_n)$.

The target residual

$$b_n(x) = \omega_n(x) - A(w_n(x))$$

is bounded on E by the common target range. Outside E it is the old residual $b\chi_R(u)$, hence is bounded and vanishes off the old compact source support. Thus all b_n vanish outside one compact set, and $\|b_n\|_\infty \leq M_*$ uniformly in n. Choose R_*, independently of n, larger than $2R$ and the target velocity bound on E, with $M_*/R_* \leq \beta/4$. At every point where $b_n \neq 0$, $|w_n| < R_*$, so

$$\omega_n = A(w_n) + b_n \chi_{R_*}(w_n). \tag{25}$$

Solve the exact cutoff problem of lemma 5 with this source, obtaining $U_n^* = (u_n^*, z_n^*)$ and the original mass and pressure data.

For comparison with the affine approximant define

$$d_n = \widetilde{\omega}_n - A(\widetilde{w}_n) - b_n \chi_{R_*}(\widetilde{w}_n).$$

Subtract (25). The direct drag difference tends to zero in L^s. The targets are the fixed rough u plus uniformly bounded increments, so their rth powers are uniformly integrable; the L^r-close affine velocities have the same property. Lemma 3 therefore makes their A difference tend to zero in L^s. The cutoff difference tends to zero in L^s by its Lipschitz bound, the common M_* and $L^r \subset L^s$ on the bounded domain. Consequently $\|d_n\|_s \to 0$.

The exact and approximate velocities lie in $\mathcal{C}$ and both drag fields equal f minus a gradient with pressure in $P + V$. Lemma 5 yields

$$\|u_n^* - \widetilde{w}_n\|_r^{r-1} \leq \frac{2}{\gamma} \|d_n\|_s \longrightarrow 0.$$

Uniform integrability passes to these exact velocities. The identity

$$z_n^* - \widetilde{\omega}_n = A(u_n^*) - A(\widetilde{w}_n) + b_n(\chi_{R_*}(u_n^*) - \chi_{R_*}(\widetilde{w}_n)) - d_n$$

and lemma 3 then give drag convergence in L^s. Thus $U_n^* - W_n \to 0$ in $L^r \times L^s$.

The H integrals also become close, despite the possible rough tails. All exact solutions have the common cutoff radius R_* and source bound M_*, so (15) bounds $H(U_n^*)$ uniformly. The targets have a uniform H bound as well: off E they are U, and on E they lie in a common compact range. They are tight in measure, being the fixed pair U plus uniformly bounded increments. Discard a small-measure large-value set of U. On its complement W_n lies in a common compact range; outside another set of vanishing measure, product-norm closeness places U_n^* in a slightly larger

range. Uniform continuity of H there proves $H(U_n^*) - H(W_n) \to 0$ in measure. Its uniform bound makes the convergence L^1: split at a fixed small magnitude as in the proof of lemma 3. Only these exact fields and ideal targets enter this argument; no integrability of $H(\widetilde{W}_n)$ is needed.

It follows that $\int_\Omega H(U_n^*) < \eta$ for all sufficiently large n. Also product-norm closeness and the bounded target increment norms imply

$$\limsup_n \left(\|u_n^* - u\|_r^r + \|z_n^* - z\|_s^s\right) \le \int_\Omega H(U) + \epsilon.$$

A sufficiently large finite n satisfies both inequalities in (24), including when the original integral is zero. Set $U' = U_n^*$. The new source and cutoff bounds may depend on this step, but were fixed throughout its frequency limit. □

Proof of theorem 1. Start with the auxiliary solution $U^{(0)} = (u^{(0)}, z^{(0)})$ from lemma 4 with $F = f$. It is a compact-source cutoff solution with $b = 0$. On $|u^{(0)}| > 1$ its defect vanishes, and on $|u^{(0)}| \le 1$ its drag is $\beta u^{(0)}$, so $\int_\Omega H(U^{(0)}) < \infty$. Apply lemma 9 to $U^{(j)}$ with $\eta_j = 2^{-r(j+1)}$. Then, for $j \ge 1$,

$$\int_\Omega H(U^{(j)}) \le 2^{-rj},$$
$$\|u^{(j+1)} - u^{(j)}\|_r^r + \|z^{(j+1)} - z^{(j)}\|_s^s \le 3\,2^{-rj}.$$

The first step has finite cost. The remaining velocity norm increments are bounded by $3^{1/r}2^{-j}$, and the drag norm increments by $3^{1/s}2^{-(r-1)j}$. Both series converge. Completeness gives

$$u^{(j)} \longrightarrow u \text{ in } L^r, \qquad z^{(j)} \longrightarrow z \text{ in } L^s.$$

Each $z^{(j)} = f - \nabla p^{(j)}$ with $p^{(j)} \in P + V$. The bounded inverse gradient of proposition 2 makes $p^{(j)} - P$ Cauchy in $W^{1,s}$, giving $p \in P + V$ and $z = f - \nabla p$. For every fixed $\psi \in V$, Holder's inequality passes the mass identity to the limit. This preserves exactly the original normal-flux functional and, in pure flux, the pressure mean.

An almost-everywhere convergent subsequence exists in the finitely many pair components by [1, Theorem 4.9]. Continuity of H and Fatou's lemma give

$$0 \le \int_\Omega H(u, z) \le \liminf_j \int_\Omega H(U^{(j)}) = 0.$$

Therefore $(u, z) \in \Gamma$ almost everywhere. In particular $|u| \ge 1$, and $z = \phi_2(|u|^2)u$ where $|u| > 1$. Define the single global $h = z\mathbf{1}_{\{|u|=1\}} \in L^s$. The original law (3) then gives $z \in \Lambda(u)$, and all conditions in (5) hold. □

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AI-use statement

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